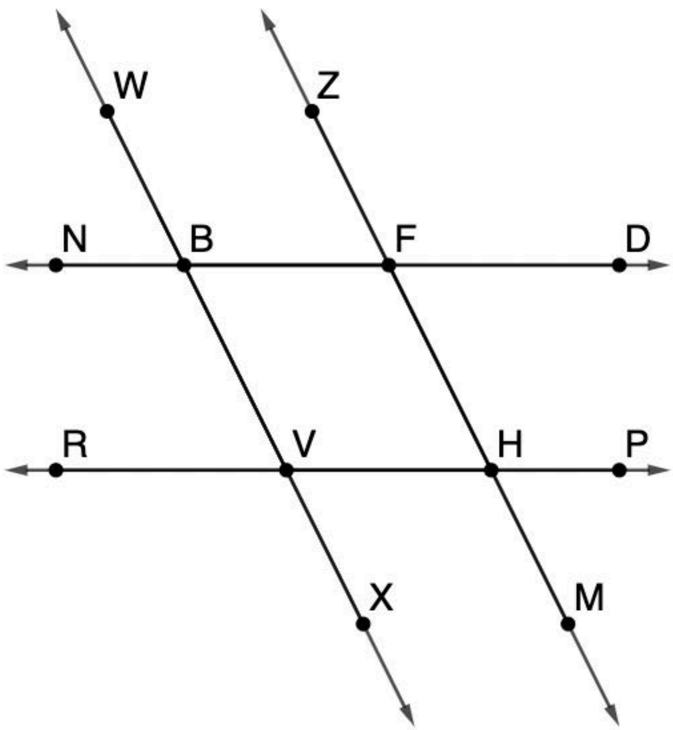
Geometry Name____________________

Unit 3 Date____________________

Lesson 8 Practice Period___________________

Lines $\overleftrightarrow{ND}$, $\overleftrightarrow{RP}$, $\overleftrightarrow{WX}$ and $\overleftrightarrow{ZM}$ are shown.



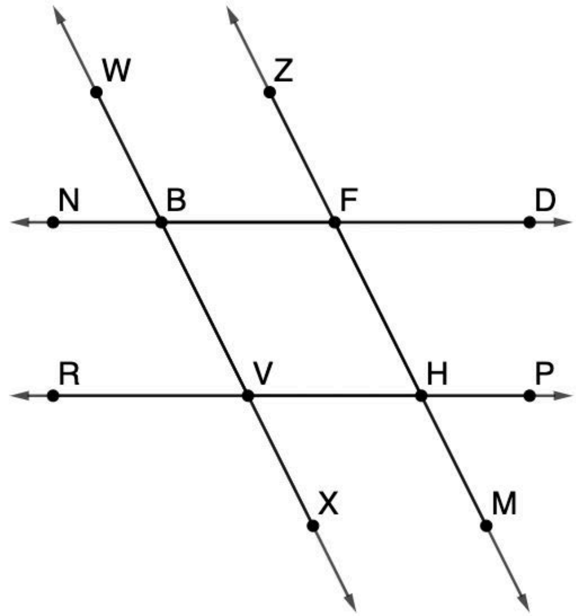
Circle the contradictory piece of information in each row.

	Item 1	Item 2	Item 3	Item 4
1.	$\angle NBW \cong \angle MHP$	$\overleftrightarrow{ND} \parallel \overleftrightarrow{RP}$	$\overleftrightarrow{WX} \nparallel \overleftrightarrow{ZM}$	$\angle BFZ \cong \angle HFD$
2.	$\angle ZFD$ and $\angle NBV$ are supplementary	$m\angle ZFD = 118^o$	$\overleftrightarrow{WX} \parallel \overleftrightarrow{ZM}$	$\angle RVX \cong \angle FHP$
3.	$\angle BFH \cong \angle HVB$	$\angle NBW$ and $\angle MHP$ are supplementary	$\angle NBV \cong \angle ZFD$	$\angle ZFD$ and $\angle MHP$ are supplementary
4.	$\angle WBN \cong \angle VBF$	$\angle RVX$ and $\angle NBW$ are supplementary	$\angle BFZ \cong \angle VHM$	$\overleftrightarrow{ND} \parallel \overleftrightarrow{RP}$
5.	$\overleftrightarrow{ND} \parallel \overleftrightarrow{RP}$	$m\angle WBF = 203^o$	$\overleftrightarrow{WX} \parallel \overleftrightarrow{ZM}$	$\angle ZFD$ and $\angle RVB$ are supplementary

Choose from the bank of statements and reasons to complete the paragraph proof.

Given: $\overleftrightarrow{WX} \parallel \overleftrightarrow{ZM}$ and $\angle ZFD \cong \angle VHM$

Prove: $\angle DFH \cong \angle BVR$



It is given that 6.____________________________. Therefore, $\angle BVR \cong \angle FHV$ because if two parallel lines are intersected by a transversal then 7.__________________________. It is also given that 8.__________________________. If 9. ________________________, then the two lines that are intersected by a transversal are parallel and it can be written that $\overleftrightarrow{ND} \parallel \overleftrightarrow{RP}$. Then, by the alternate interior angles theorem, it must be true that 10. _________________. Therefore, by transitive property of congruence, $\angle DFH \cong \angle BVR$.

Statements and Reasons	
$\overleftrightarrow{WX} \parallel \overleftrightarrow{ZM}$	corresponding angles are congruent
$\angle DFH \cong \angle BVR$	a linear pair is supplementary
$\angle DFH \cong \angle FHV$	vertical angles are congruent
$\overleftrightarrow{ND} \parallel \overleftrightarrow{WX}$	alternate interior angles are congruent
$\angle WBF$ and $\angle NBV$	alternate exterior angles are congruent
$\angle ZFD \cong \angle VHM$	transitive property of congruence